

COVARIATE SHIFT COMPOUNDING ERROR FLYWHEEL & RECOVERY DYNAMICS

Quadratic Divergence $O(T^2\epsilon)$ in Naive Behavioral Cloning vs Linear Bounded Tube $O(T\epsilon)$ in DAgger Corrective Aggregation

× NAIVE BEHAVIORAL CLONING (OPEN-LOOP DRIFT FLYWHEEL)

Error: $O(T^2\epsilon)$

1. Single-Step Perturbation ϵ

Friction / torque / sensor noise

2. State Exits Support

$s_{t+1} \notin \text{supp}(d^{\pi^*})$

4. Amplified Deviation

Pushed further off-manifold

3. Arbitrary Action $\hat{a}$

Zero constraint from ERM loss

Mathematical Compounding Mechanism:

$$E[\text{Error}_{BC}] \leq \sum t \cdot \epsilon = [T(T+1)/2] \cdot \epsilon = O(T^2\epsilon)$$

Example ($T = 500$ steps, $\epsilon = 0.01$) $\Rightarrow$ Cumulative Error Factor = 2,500×

✓ CORRECTIVE AGGREGATION / DAgger (STABILIZING FLYWHEEL)

Error: $O(T\epsilon)$

1. Policy Visits $s \sim d_{\text{policy}}$

Learner visits perturbed state

2. Expert Labels $\pi^*(s)$

Recovery action demonstrated

4. Closed-Loop Recovery

Restoring torque restores state

3. Aggregate to Dataset D

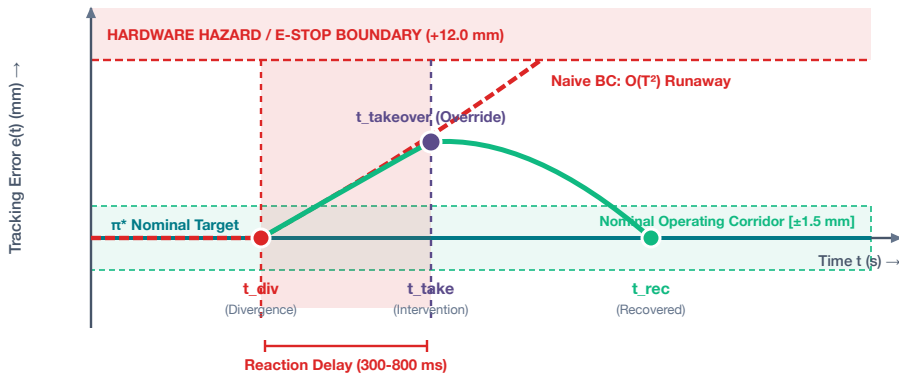
$\text{supp}(D) \supseteq \text{recovery corridor}$

Mathematical Linear Bounded Corridor:

$$E[\text{Error}_{CA}] \leq T \cdot \epsilon = O(T\epsilon) \text{ [Ross \& Bagnell 2011]}$$

Example ($T = 500$ steps, $\epsilon = 0.01$) $\Rightarrow$ Cumulative Error Factor = 5.0× (500× reduction!)

TRAJECTORY DIVERGENCE & INTERVENTION CUT BOUNDARY



THE 3 DATA CURATION LAWS FOR PHYSICAL INTERVENTIONS:

- 1. Truncate Autonomous Rollout at t_{div} (NOT t_{takeover}):**
Cut rollout where policy drifted. Prevents neural networks from learning that approaching a hazard precedes goal completion.
- 2. Discard Corrupted Drift Interval $[t_{\text{div}}, t_{\text{takeover}}]$:**
Purge 30-80 frames logged during human reaction delay. These transitions contain divergent commands falsely logged as nominal.
- 3. Ingest Recovery Demonstration $[t_{\text{takeover}}, t_{\text{rec}}]$ into DAgger Buffer:**
Treat rescue as expert feedback on policy distribution d_{policy} , expanding support to recovery corridor and bounding error to $O(T\epsilon)$.